

THE SOUND THAT LEAKS YOUR LOCATION

Aircraft-Doppler geolocation - one-page technique sheet

Source technique: colsto, Nov 2026 | Data: public ADS-B | W1D0WM4K3R // B1SCU1TK1D.com

November 4, 2026 - concept and capability sheet, free to share

THE CLAIM

- **In one line:** Any recording with ambient audio - a phone video, a security clip, a call - carries an inaudible fingerprint of every propeller aircraft that flew over during it. That fingerprint can place the recording to within a few hundred feet.
- **The signature:** A flyover imprints a Doppler curve between 30-250 Hz: the aircraft's engine frequency, shifted up on approach and down on departure. Inaudible to ears, invisible to casual listening, plain as day in a spectrogram.

HOW IT WORKS - THE FIVE-STEP PIPELINE

- **1. Spectrogram:** Compute the time-frequency map of the audio. Boost the low band until the faint curves appear.
- **2. Measure:** Each curve encodes velocity (size of the frequency shift), slant distance (steepness of the S), and the exact time the plane was closest (CPA).
- **3. Fetch truth:** ADS-B broadcasts every plane's position, speed, altitude, and time. Public archives hold every flight over the US for any given day.
- **4. Match:** Model what each candidate point on the ground would have heard - real flight geometry, sound-travel delay, terrain elevation - and score it against the recording.
- **5. Lock:** Two flyovers with a measured time gap, or one flyover with a known recording time, constrains the search to one place. The offset between two events is locked by geometry - measurement error cannot fake a match.

WHAT THIS MEANS

- **For OSINT:** Footage with zero visual clues becomes locatable. The sound is everywhere, and the data to match it is public.
- **For OPSEC:** It survives phone-call compression, walls, and loud foreground audio. High-pass filters fail. Only cuts in the audio - or active noise removal - break the chain. Any footage published from a sensitive site should carry no raw ambient audio.
- **The honest limit:** This is an advanced, computationally heavy technique - months of tooling upstream. The odds of an average person being targeted this way are slim; there are a hundred easier ways to find someone. Its real value: forensics on audio with nothing else to go on.